Aniket Deshpande

University of Illinois, Urbana-Champaign

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Areas of Research

Many-Body Physics · Machine Learning

Education

2023–2027 **University of Illinois, Urbana-Champaign**
B.S. in Physics, *Specialization in Physics & Computation*

Publications & Preprints

2026 **Fermionic Shadow Regression: Amortized Learning of Molecular Electron Dynamics** — In preparation
A. Deshpande, L. Mantilla Calderón, K. Panicker, M. Bagherimehrab, A. Aspuru-Guzik.

2026 **Feature-Resolved Attention** — ICML Mech. Interp. Workshop
D. Manning-Coe, N. Aljaafari, J. Stephenson, K. Kapre, H. Jain, **A. Deshpande**.

2026 **Crosscoding Through Time: Sparse Feature Discovery Across Sequence Positions** — ICML Mech. Interp. Workshop
D. Manning-Coe, H. Xuanyuan, **A. Deshpande**, A. Shportko, W. Fei.

Research

Summer 2026 **Lawrence Berkeley National Lab** — *Berkeley, CA*
Student Researcher, *Mentor: Prof. Roel Van Beeumen*
Incoming summer 2026 — randomized algorithms for isometric tensor networks.

Feb. 2026– **Supervised Program for Alignment Research** — *Remote*
Research Fellow
Developing temporal crosscoders, a sparse dictionary learning architecture that ties decoder atoms across token positions to exploit multi-position temporal structure in transformer residual stream activations. Evaluating against standard SAEs and layer-wise crosscoders on sparse probing benchmarks and downstream reasoning-steering pipelines.

Feb.–May 2026 **Vector Institute for Artificial Intelligence** — *Toronto, ON*
Research Intern, *PI: Prof. Alán Aspuru-Guzik*
Developing geometry-conditioned regressors for fermionic classical shadow data to predict time-resolved molecular observables, with accompanying sample-complexity and time-resolution guarantees.

Sep. 2024– **Lab for Parallel Numerical Algorithms** — *Urbana, IL*
Undergraduate Researcher, *PI: Prof. Edgar Solomonik*
Developing an annealed importance sampling algorithm with Glauber dynamics for approximating tensor network contractions, with provable mixing time guarantees and applications to quantum simulation. Proved that alternating Mahalanobis distance minimization for CP decomposition performs maximum likelihood estimation under a Kronecker-structured noise covariance model.

Industry Experience

May–Aug. 2024 **Space Dynamics Laboratory** *Albuquerque, NM*
Ionospheric Analyst Intern
Developed a Python scraper to expedite data collection of NICT ionograms (600+/hour). Implemented and benchmarked noise-reduction filters in Python and Julia (PSNR, MSE, SSIM). Researched CNN architectures for automatic ionogram scaling.

Talks & Posters

Aug. 2025 [P] "Approximating Tensor Contractions with Annealed Importance Sampling." *QSim 2025*, New York, NY.

Dec. 2024 [P] "Quantum Circuit Volume for Graph Models." *Illinois Mathematics Lab Open House*, Urbana, IL.

Selected Training

Aug.–Dec. 2025 Independent Study, Information Geometry, with Prof. Matthew Singh

Aug. 2025 QSim Summer School — NSF RQS (hosted at IBM, NYC)

May 2025 UQ & ML for Physical Systems — IMSI, University of Chicago

Service & Affiliations

May 2025– Membership Director, SIAM @ University of Illinois, Urbana-Champaign

May 2025– Member, Society for Industrial & Applied Mathematics

Coursework

Statistical Mechanics, Quantum Information Theory^{G}, Electromagnetism I, Classical Mechanics I & II, Special Relativity, Real Analysis, High-Dimensional Statistics^{AG}, Optimal Control Theory^{AG}, Representation Theory^{AG}, Stochastic Processes, Deep Learning Theory^{G}, Theoretical Neuroscience^{G}, Biologically-Plausible AI^{G}, Machine Learning, Deep Generative Models, Data Structures & Algorithms.

^{G} graduate coursework ^{A} audited

Skills

Languages: Python, C/C++, Java, Julia, Mathematica.

Libraries: NumPy, SciPy, Pandas, Matplotlib, PyTorch, SymPy.

Tools: Git, LaTeX, Conda, Shell, Jupyter.

Last updated: June 14, 2026